RWorksheet_prado#2

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```r
# 1a. Sequence from 5 to 5
5:5
```

```
## [1] 5
```

```r
# 1b. Value of x <- 1:7
x <- 1:7
x
```

```
## [1] 1 2 3 4 5 6 7
```

```r
# 2a. seq(1, 3, by=0.2)
seq(1, 3, by = 0.2)
```

```
##  [1] 1.0 1.2 1.4 1.6 1.8 2.0 2.2 2.4 2.6 2.8 3.0
```

```r
# 3. Ages of 50 workers
ages_50 <- c(34, 28, 22, 36, 27, 18, 52, 39, 42, 29, 35, 31, 27, 22, 37, 34, 19, 20, 57, 49, 50, 53, 41

# 3a. Length/Data Points
length(ages_50)
```

```
## [1] 50
```

```r
# 3b. Descriptive Statistics
mean(ages_50)
```

```
## [1] 35.74
```

```r
median(ages_50)
```

```
## [1] 36.5
```

```r
min(ages_50)
```

```
## [1] 17
```

```r
max(ages_50)
```

```
## [1] 57
```

```r
# Use table() to find the mode (most frequent value)
table(ages_50)
```

```
## ages_50
## 17 18 19 20 22 24 25 27 28 29 31 33 34 35 36 37 38 39 40 41 42 44 46 48 49 50
##  1  2  2  2  3  1  1  3  1  1  1  1  3  2  1  4  1  1  1  3  2  1  1  1  2  2
## 51 52 53 57
##  1  1  2  2
```

```r
# 6a. Create Data Frame
month <- c("Jan", "Feb", "March", "Apr", "May", "June")
price_per_liter <- c(52.50, 57.25, 60.00, 65.00, 74.25, 76.50)
purchase_quantity <- c(25, 30, 10, 15, 40, 8)
fuel_data <- data.frame(Month = month, Price_PHP = price_per_liter, Quantity_Ltr = purchase_quantity)
fuel_data
```

```
##    Month Price_PHP Quantity_Ltr
## 1    Jan     52.50           25
## 2    Feb     57.25           30
## 3 March      60.00           10
## 4    Apr     65.00           15
## 5    May     74.25           40
## 6   June     76.50            8
```

```r
# 6b. Average Fuel Expenditure (Weighted Mean)
weighted.mean(fuel_data$Price_PHP, fuel_data$Quantity_Ltr)
```

```
## [1] 63.96094
```

```r
Celebrity_Name <- c(
  "Tom Cruise", "Rolling Stones", "Oprah Winfrey", "U2", "Tiger Woods",
  "Steven Spielberg", "Howard Stern", "50 Cent", "Cast of the Sopranos", "Dan Brown",
  "Bruce Springsteen", "Donald Trump", "Muhammad Ali", "Paul McCartney", "George Lucas",
  "Elton John", "David Letterman", "Phil Mickelson", "J.K Rowling", "Bradd Pitt",
  "Peter Jackson", "Dr. Phil McGraw", "Jay Lenon", "Celine Dion", "Kobe Bryant"
)

Power_Ranking <- 1:25
Annual_Pay <- c(
  67, 90, 225, 110, 90, 332, 302, 41, 52, 88,
  55, 44, 55, 40, 233, 34, 40, 47, 75, 25,
  39, 45, 32, 40, 31
)
```

```r
JK_Rowling_Index <- which(Celebrity_Name == "J.K Rowling")

Power_Ranking[JK_Rowling_Index] <- 15

Annual_Pay[JK_Rowling_Index] <- 90

Forbes_Data_Modified <- data.frame(
  Ranking = Power_Ranking,
  Celebrity = Celebrity_Name,
  Pay_Million_USD = Annual_Pay
)

cat("Modified data for J.K. Rowling (Row ", JK_Rowling_Index, "):", "\n")
```

```
## Modified data for J.K. Rowling (Row  19 ):
```

```r
Forbes_Data_Modified[JK_Rowling_Index, ]
```

```
##    Ranking   Celebrity Pay_Million_USD
## 19      15 J.K Rowling              90
```